

CLOUD SECURITY ASSESSMENT REPORT

AWS EC2 Security Hardening & Monitoring Lab

Executive Summary

This report documents the deployment, security hardening, monitoring, and validation of an Amazon Web Services (AWS) Elastic Compute Cloud (EC2) environment. The objective of the assessment was to implement cloud security best practices to reduce infrastructure risk, strengthen access controls, and establish monitoring capabilities.

The project focused on securing an Amazon Linux 2023 EC2 instance through network security controls, SSH hardening, user privilege management, and CloudWatch-based monitoring. Following implementation, all cloud resources were reviewed and securely decommissioned to prevent unnecessary costs and maintain proper cloud resource governance.

Section 1: Cloud Environment Identification

Cloud Platform

- Cloud Provider: Amazon Web Services (AWS)
- Service: Amazon EC2
- Region: ap-south-1 (Mumbai)
- Operating System: Amazon Linux 2023
- Instance Type: t3.micro

Security Services Used

- AWS EC2
- AWS Security Groups
- SSH Key Pair Authentication
- Amazon CloudWatch
- Amazon SNS
- Linux Access Controls

Assessment Objective

The primary objective of this project was to design and implement a secure cloud environment that follows industry security practices including:

- Restrictive network access
 - Secure remote administration
 - Host hardening
 - Least-privilege access control
 - Security monitoring
 - Alert generation
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Section 2: Environment Deployment & Investigation Activities

The cloud security assessment began with the deployment of an Amazon EC2 instance running Amazon Linux 2023. During deployment, a dedicated Security Group was created to control inbound network access.

To reduce exposure to external threats, Secure Shell (SSH) access was restricted exclusively to the administrator's public IP address. This configuration prevented unrestricted internet access to the server and reduced the attack surface exposed to potential threat actors.

Following successful deployment, the instance was accessed using SSH key-based authentication. Password-based authentication was not used during deployment, reducing the risk associated with credential-based attacks.

The operating system was updated using the latest available package repositories to ensure all known security patches were installed. A dedicated administrative account named "cloudadmin" was created to separate administrative operations from default system accounts.

The SSH daemon configuration was then reviewed and hardened. Root login was disabled, password authentication was disabled, maximum authentication attempts were limited, and unnecessary SSH functionality was removed. These controls were implemented to reduce brute-force attack opportunities and improve access security.

After hardening activities were completed, AWS CloudWatch monitoring was configured. A CPU utilization alarm threshold was established, and SNS email notifications were enabled to provide proactive visibility into infrastructure events.

The environment was continuously reviewed to validate successful implementation of all security controls and monitoring functions.

Section 3: Security Controls Implemented

Identity and Access Security

Implemented Controls:

- SSH Key-Based Authentication
- Dedicated Administrative User
- Principle of Least Privilege
- Administrative Role Separation
- Restricted Privileged Access

Security Benefit:

These controls reduce the likelihood of unauthorized access and improve accountability by separating administrative activities from default accounts.

Network Security

Implemented Controls:

- Security Group Firewall Configuration
- SSH Access Restricted to Trusted IP Address
- Inbound Traffic Restriction
- Reduced External Exposure

Security Benefit:

Restricting SSH access significantly reduces the attack surface available to external threat actors and minimizes the risk of unauthorized access attempts.

Host Security

Implemented Controls:

- Operating System Updates
- Security Patch Management
- SSH Hardening
- Root Login Disabled
- Password Authentication Disabled
- Authentication Attempt Restrictions

Security Benefit:

These controls reduce vulnerabilities associated with outdated software, weak authentication methods, and brute-force attack techniques.

Monitoring and Detection

Implemented Controls:

- CloudWatch Monitoring
- CPU Utilization Alerts
- SNS Email Notifications
- Infrastructure Visibility

Security Benefit:

Continuous monitoring provides visibility into resource behavior and enables rapid detection of unusual activity requiring administrative attention.

Section 4: Security Findings

Finding 1: Default Configuration Risk

Risk Level: Medium

Observation:

Default cloud deployments often provide configurations that may expose administrative services to broader network access than required.

Remediation:

Implemented restrictive Security Group policies limiting SSH access to a trusted source IP.

Status:

Remediated

Finding 2: Credential-Based Authentication Risk

Risk Level: High

Observation:

Password-based authentication can increase exposure to credential stuffing and brute-force attacks.

Remediation:

Implemented SSH key authentication and disabled password-based logins.

Status:

Remediated

Finding 3: Excessive Privilege Risk

Risk Level: Medium

Observation:

Using a single administrative account for all operations can reduce accountability and increase risk.

Remediation:

Created a dedicated administrative user and implemented least-privilege principles.

Status:

Remediated

Finding 4: Monitoring Visibility Gap

Risk Level: Medium

Observation:

Infrastructure without monitoring lacks visibility into operational anomalies.

Remediation:

Configured CloudWatch monitoring and SNS notifications.

Status:

Remediated

Section 5: Risk Assessment

Risk Category	Severity	Status
Unauthorized Remote Access	High	Mitigated
Brute Force Attacks	High	Mitigated
Weak Authentication	High	Mitigated
Excessive Privileges	Medium	Mitigated
Lack of Monitoring	Medium	Mitigated

Section 6: Recommendations

The following improvements are recommended for future security maturity:

1. Implement AWS IAM Roles and Policies.
2. Enable AWS CloudTrail logging.
3. Deploy AWS GuardDuty threat detection.
4. Configure AWS Config compliance monitoring.
5. Implement automated infrastructure deployment using Terraform.
6. Integrate vulnerability scanning and compliance assessments.
7. Enable centralized logging and security analytics.

Section 7: Conclusion

The AWS EC2 Security Hardening & Monitoring Lab successfully demonstrated foundational cloud security engineering practices within AWS. Security controls were implemented across identity management, network security, host security, and monitoring domains.

The project validated the ability to securely deploy cloud infrastructure, implement access control mechanisms, harden Linux systems, establish monitoring workflows, and manage cloud resources according to security best practices.

The environment was successfully deployed, secured, monitored, documented, and decommissioned, demonstrating practical cloud security skills relevant to entry-level Cloud Security Engineer, Cloud SOC Analyst, and Cloud Operations Security roles.